Threshold Optimization

March 25, 2026

When detecting ties (TPs) is more important than not detecting ties (FNs) or vice versa, accuracy, which equally weighs TPs and FNs, is not the ideal metric for comparing algorithm performance. Therefore, in addition to evaluating algorithms at a .50 cut-off, we optimized two performance metrics (F_β at $\beta = 1$ and $\beta = 2$) to identify additional cut-off values.

1 Method

We used 10 fold cross validation to optimize thresholds for each algorithm using a range of 101 candidate values between 0 and 1 with .01 increments to convert predicted probabilities into binary classifications. The trained model was refitted using observations in the training fold and performance metrics were calculated on observations in the validation fold. The folds were created using stratified sampling to ensure similar tie proportions in each fold.

We found the optimal cut-off that corresponded to maximum value of six metrics: F1 score, normalized Matthews Correlation Coefficient (nMCC), Youden's J (Youden, 1950), sensitivity, two weighted F scores with $F_{\beta=2}$ (that weighed recall twice as important than precision) and $F_{\beta=10}$ (that weighed recall 10 times as important than precision).

1.1 Metrics

Sensitivity, the metric we care most about, always achieves its maximum at the lowest specified cut-off (i.e., by classifying nearly everything as a tie) but a classifier like that is not very useful.

$$Sensitivity = \frac{TP}{TP + FN}$$

Matthews correlation coefficient (MCC) corresponds to calculating the Pearson r correlation between observed and predicted binary classifications. High scores indicate that prediction correctly classified substantial non-ties as well as substantial ties, in both balanced and imbalanced data.

$$MCC = \frac{(TP \times TN - FP \times FN)}{\sqrt{((TP + FP)(TP + FN)(TN + FP)(TN + FN)}}$$

It ranges form -1 to 1 with 0 meaning prediction quality is the same as what would be expected by random chance. To make it's scale comparable to other metrics, we used normalized MCC,

$$nMCC = \frac{MCC + 1}{2}$$

Another classification metric is Youden's j (Youden, 1950) which is supposed to penalize FPs and FNs equally across all class proportions, but is in reality dependent on the prevalence rate of the outcome if the outcome isn't balanced (Smits, 2010). The cost ratio of this metric is not fixed, but corresponds to the population prevalence rate, so thresholds will change according to populations.

$$J = sensitivty + specificity - 1$$

Weighted F score: if β describes the ratio of importance of recall to precision, then

$$F_\beta = (1+\beta^2)\frac{(precision \times recall)}{\beta^2(precision + recall)}$$

If $\beta = 1$, then recall and precision are equally important and the formula reduces to the standard F-1 score. If $\beta > 1$, then recall is more important than precision, and if $\beta < 1$, then precision is more important than recall.

We note that F-1 would have been inappropriate if TNs were more important to detect, as it would decrease rapidly if we classify more ties as positive. This is because F-1 scores suffer from two key limitations– asymmetry and dependence on imbalance (Yedidia, 2016). In our context, our models are predicting mostly negative cases and our dataset is composed of mostly negative instances, so F-1 score and MCC are both informative indices (Hand & Christen, 2018, e.g.,).

Smits et al., 2007 also demonstrate setting custom cut-off scores not based on optimizing metrics alone, but by defining costs and benefits in a decision theory framework to quantify the subjectivity of costs based on substantive knowledge.

1.2 Combining results

As expected, optimizing sensitivity, Youden's J, or $F_\beta = 10$ led to suggested cut-offs that were too liberal. nMCC was a proxy for accuracy as it equally penalized FPs and FNs, so the cut-offs were more conservative. F-1 and $F_\beta = 2$ based cut-offs fell mid-way. We can see that regularized regression approaches had poorer prediction quality across thresholds; their plots have a different scale than the others cause the optimal cut-offs were very low. Although six metrics were optimized in this process, we retained F1 score and $F_{\beta=2}$ due to our interest in preferentially weighing sensitivity while retaining precision. We decided to aggregate cross validation results across ensemble methods to arrive at one cut-off to be applied to all algorithms, .29 when optimizing $F_{\beta=1}$ and .12 when optimizing $F_{\beta=2}$.

References

Hand, D., & Christen, P. (2018). A note on using the f-measure for evaluating record linkage algorithms. *Statistics and Computing*, *28*(3), 539–547.

Smits, N. (2010). A note on youden's j and its cost ratio. *BMC Med Res Methodol*, *10*, 89.

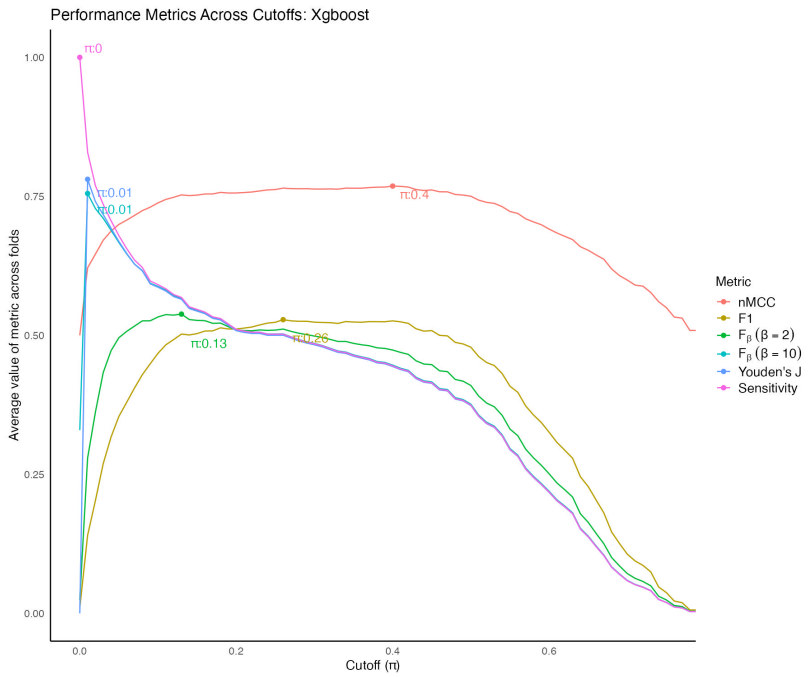
Smits, N., Smit, F., Cuijpers, P., & De Graaf, R. (2007). Using decision theory to derive optimal cut-off scores of screening instruments: An illustration explicating costs and benefits of mental health screening. *International Journal of Methods in Psychiatric Research*, *16*(4), 219–229.

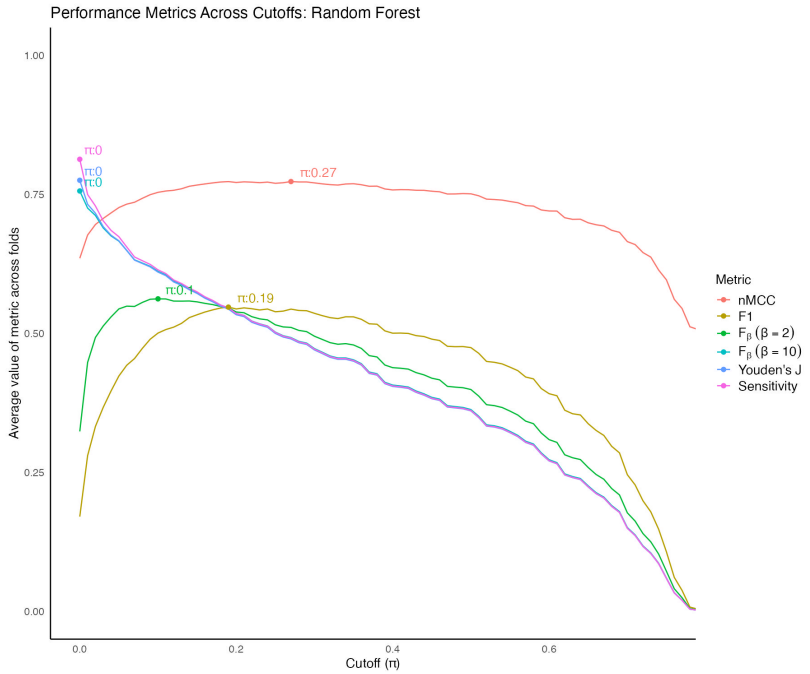
Yedidia, A. (2016, December 8). *Against the f-score*. https://adamyedidia.files.wordpress.com/2014/11/f_score.pdf

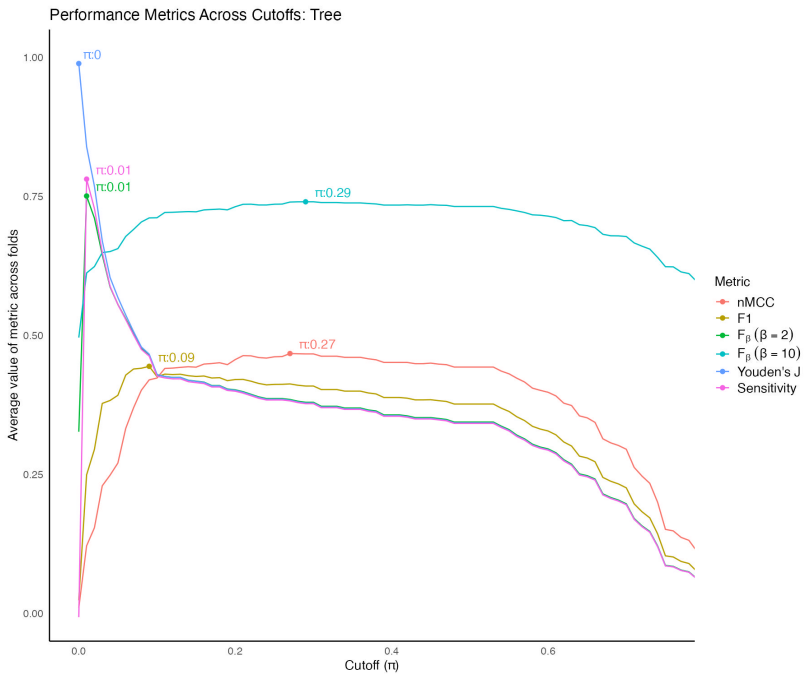
Youden, W. J. (1950). Index for rating diagnostic tests. *Cancer*, *3*(1), 32–35. https://doi.org/https://doi.org/10.1002/1097-0142(1950)3:1⟨32::AID-CNCR2820030106⟩3.0.CO;2-3

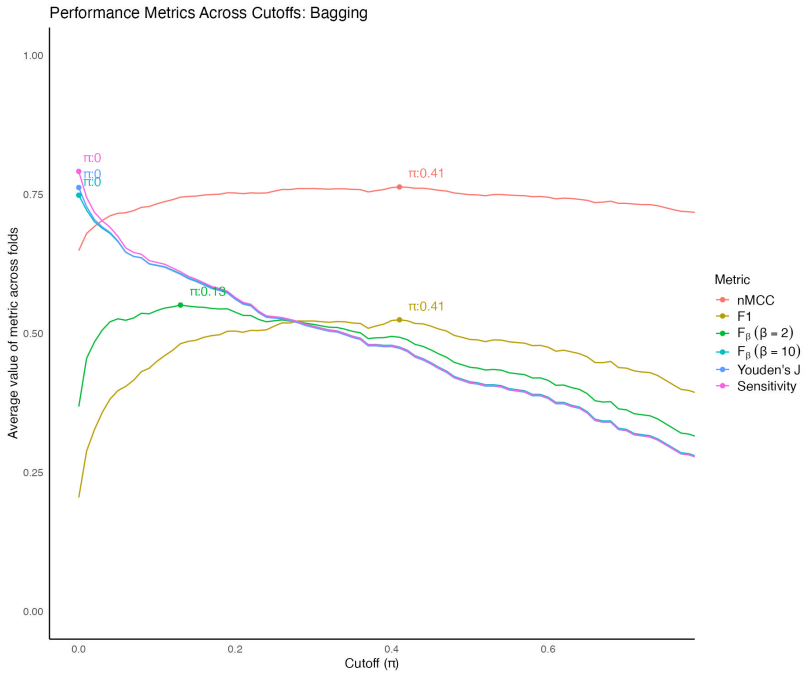
2 Plots

2.1 Tree based methods

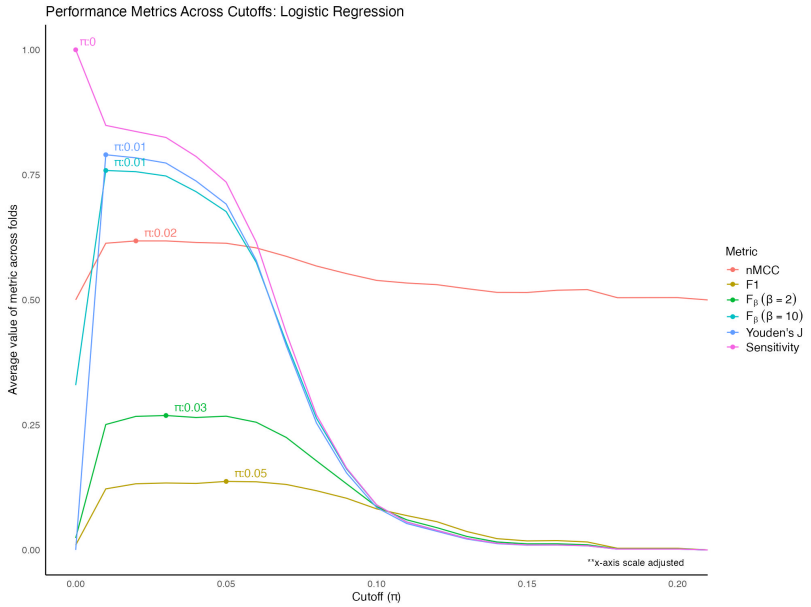

Performance Metrics Across Cutoffs: Xgboost
Average value of metric across folds
Cutoff (π)
1.00
0.75
0.50
0.25
0.00
0.0
0.2
0.4
0.6
π:0
π:0.01
π:0.01
π:0.4
π:0.13
π:0.26
Metric
nMCC
F1
Fβ (β = 2)
Fβ (β = 10)
Youden's J
Sensitivity


Performance Metrics Across Cutoffs: Random Forest
Average value of metric across folds
Cutoff (π)
1.00
0.75
0.50
0.25
0.00
0.0
0.2
0.4
0.6
π:0
π:0
π:0
π:0.27
π:0.1
π:0.19
Metric
nMCC
F1
Fβ (β = 2)
Fβ (β = 10)
Youden's J
Sensitivity

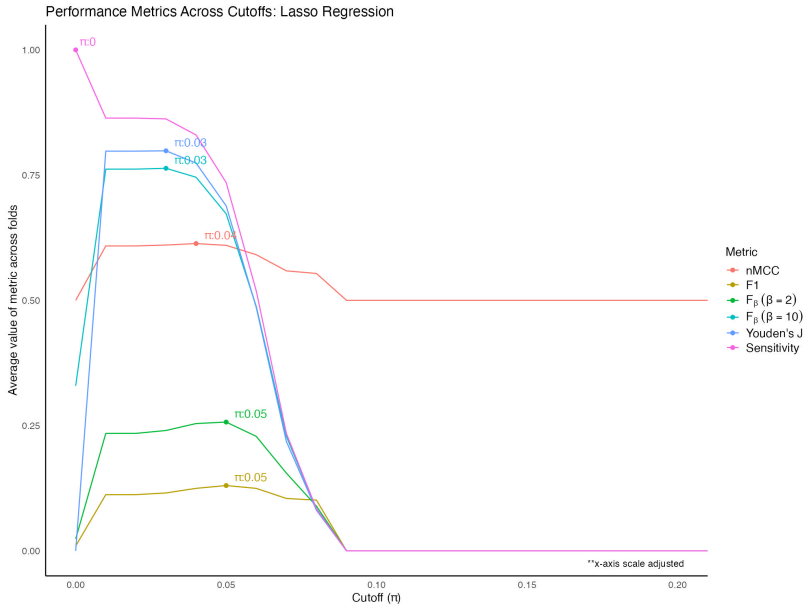

Performance Metrics Across Cutoffs: Tree
Average value of metric across folds
Cutoff (π)
π:0
π:0.01
π:0.01
π:0.29
π:0.09
π:0.27
Metric
nMCC
F1
Fβ (β = 2)
Fβ (β = 10)
Youden's J
Sensitivity

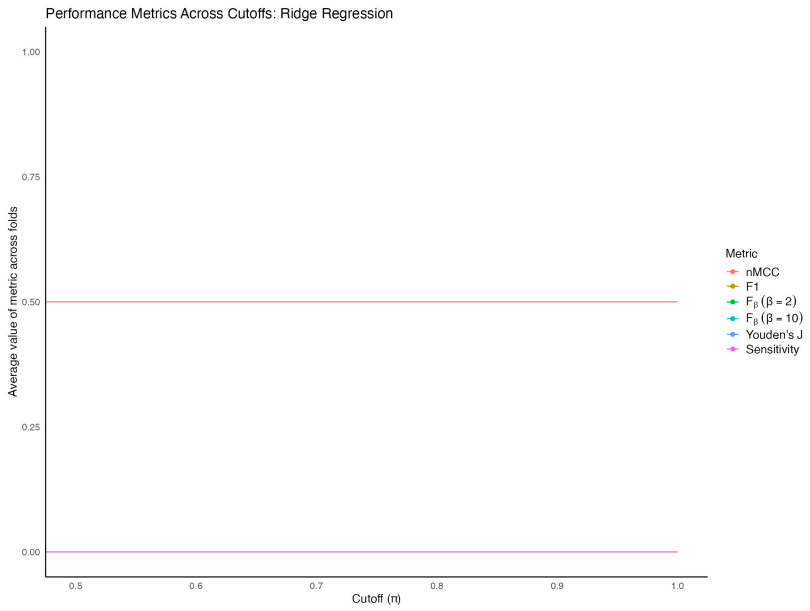

Performance Metrics Across Cutoffs: Bagging
Average value of metric across folds
Cutoff (π)
π:0
π:0
π:0
π:0.41
π:0.13
π:0.41
Metric
nMCC
F1
Fβ (β = 2)
Fβ (β = 10)
Youden's J
Sensitivity

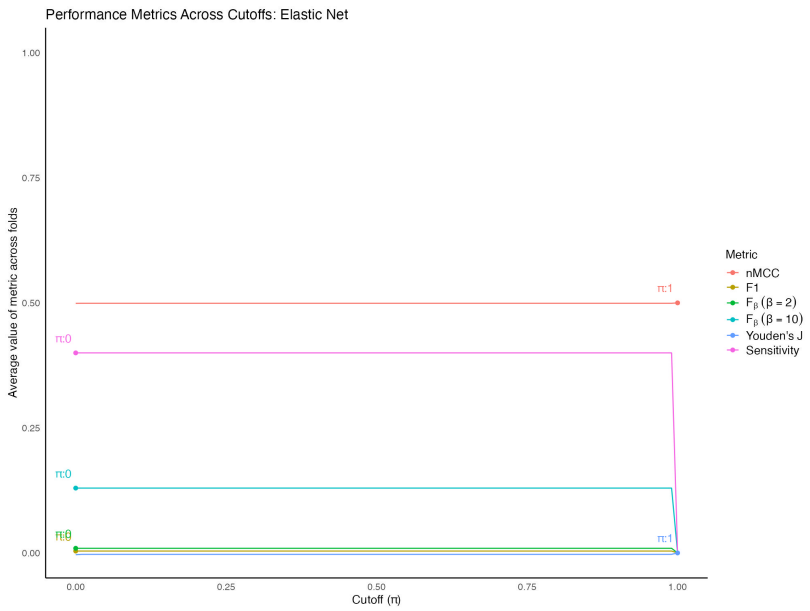
2.2 Logistic Regression



2.3 Regularized Regression methods




Performance Metrics Across Cutoffs: Ridge Regression
Average value of metric across folds
1.00
0.75
0.50
0.25
0.00
0.5
0.6
0.7
0.8
0.9
1.0
Cutoff (π)
Metric
nMCC
F1
F_β (β = 2)
F_β (β = 10)
Youden's J
Sensitivity


Performance Metrics Across Cutoffs: Elastic Net
Average value of metric across folds
1.00
0.75
0.50
0.25
0.00
π:1
π:0
π:0
π:0
π:1
0.00
0.25
0.50
0.75
1.00
Cutoff (π)
Metric
nMCC
F1
F_β (β = 2)
F_β (β = 10)
Youden's J
Sensitivity